ADVANCED CALCULUS

A Comprehensive Treatise on Differential Equations

Ordinary, Partial, and Higher-Order Linear Systems

Academic Lecture Notes & Theoretical Reference

Typeset with Publication-Quality LaTeX, TikZ, and AMS Environments

Contents

Chapter 1

Fundamentals of Differential Equations

1.1 Definitions and Categorization

Differential equations form the foundational language of dynamic modeling across pure mathematics, physics, engineering, and mathematical biology.

Definition: Differential Equation (DE)

A **differential equation** is any equation involving one or more dependent variables, one or more independent variables, and the ordinary or partial derivatives of the dependent variables with respect to the independent variables.

Differential equations are broadly classified into two main categories based on the structure of their derivatives:

Definition: Ordinary Differential Equation (ODE)

An **ordinary differential equation (ODE)** is a differential equation containing ordinary derivatives of one or more dependent variables with respect to a *single* independent variable.

Definition: Partial Differential Equation (PDE)

A **partial differential equation (PDE)** is a differential equation containing partial derivatives of one or more dependent variables with respect to *two or more* independent variables.

Classification of Differential Equations by Category

Ordinary Differential Equations (ODE):

$$\frac{d^2y}{dx^2} + xy\left(\frac{dy}{dx}\right)^2 = 0 \tag{1.1}$$

$$\frac{d^4x}{dt^4} + 5\frac{d^2x}{dt^2} + 3x = \sin(t) \tag{1.2}$$

Partial Differential Equations (PDE):

$$\frac{\partial v}{\partial \sigma} + \frac{\partial v}{\partial t} = v \tag{1.3}$$

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} = 0 \quad \text{(Laplace's Equation)} \tag{1.4}$$

Standard Calculus Derivatives (Not DEs):

$$\frac{d}{dx}(e^{ax}) = ae^{ax}, \qquad \frac{d}{dx}(uv) = u\frac{dv}{dx} + v\frac{du}{dx} \tag{1.5}$$

1.2 Order and Degree of a Differential Equation

Definition: Order of a Differential Equation

The **order** of a differential equation is defined as the order of the highest-order derivative appearing in the equation.

Definition: Degree of a Differential Equation

The **degree** of a differential equation is the algebraic degree (highest polynomial power) of the highest-order derivative present in the equation, after the equation has been cleared of radicals and fractional exponents with respect to derivatives.

Examples of Order and Degree Classification

- **1st Order, 1st Degree ODE:**
$$\frac{dy}{dx} + y = e^x$$
- **2nd Order, 1st Degree ODE:**
$$\frac{d^2y}{dx^2} + 3\frac{dy}{dx} + 2y = 0$$
- **2nd Order, 2nd Degree ODE:**
$$\left(\frac{d^2y}{dx^2}\right)^2 + y\frac{dy}{dx} = 0$$
- **1st Order, 1st Degree PDE:**
$$\frac{\partial u}{\partial t} + \frac{\partial u}{\partial x} = 0$$
- **2nd Order, 1st Degree PDE (Wave Equation):**
$$\frac{\partial^2 u}{\partial t^2} - c^2\frac{\partial^2 u}{\partial x^2} = 0$$
- **2nd Order, 2nd Degree PDE:**
$$\left(\frac{\partial^2 u}{\partial x^2}\right)^2 + \left(\frac{\partial^2 u}{\partial y^2}\right)^2 = 0$$

1.3 Linearity Criteria and Classification

Linearity plays a fundamental role in solution existence, superposition properties, and analytical tractability.

Definition: Linear Ordinary Differential Equation

An n-th order ordinary differential equation in the dependent variable y and independent variable x is called **linear** if it can be written in the explicit representation:

$$a_0(x)\frac{d^ny}{dx^n} + a_1(x)\frac{d^{n-1}y}{dx^{n-1}} + \cdots + a_{n-1}(x)\frac{dy}{dx} + a_n(x)y = f(x) \tag{1.6}$$

or in compact notation:

$$a_0(x)y^{(n)} + a_1(x)y^{(n-1)} + \cdots + a_n(x)y = f(x) \tag{1.7}$$

where $a_0(x) \neq 0$ and $a_0(x), a_1(x), \ldots, a_n(x), f(x)$ are given functions of x.

Key Criteria for Linearity

An ODE is strictly **linear** if and only if it satisfies three structural rules:

1. The dependent variable y and all its derivatives $y', y'', \ldots, y^{(n)}$ occur to the **first degree only**.
2. No product terms involving y and its derivatives (e.g., $y \cdot \frac{dy}{dx}$) or products of derivatives exist.
3. No non-linear transcendental functions of y or its derivatives are present (e.g., $\sin(y)$, e^y, $\ln(y')$).

Definition: Nonlinear Ordinary Differential Equation

An ordinary differential equation is classified as **nonlinear** if it violates any of the three linearity criteria outlined above. The general implicit form of an n-th order nonlinear ODE is:

$$F\left(x, y, \frac{dy}{dx}, \frac{d^2y}{dx^2}, \ldots, \frac{d^ny}{dx^n}\right) = 0 \tag{1.8}$$

Worked Example: Comparing Linear and Nonlinear ODEs

Linear ODE with Constant Coefficients:

$$\frac{d^2y}{dx^2} + 3\frac{dy}{dx} + 2y = 0$$

Linear ODE with Variable Coefficients:

$$\frac{d^4y}{dx^4} + x^{30}\frac{dy}{dx} + 2y = xe^x$$

Nonlinear ODE Examples:

$$\left(\frac{d^2y}{dx^2}\right)^2 + y\frac{dy}{dx} = 0 \quad \text{(Violates degree and product rule)}$$

$$\frac{dy}{dx} + \sin(y) = 0 \quad \text{(Violates transcendental function rule)}$$

$$A\frac{d^2y}{dx^2} + K\frac{d^3y}{dx^3} + x\frac{dy}{dx} = xe^x\frac{dx}{dy} \quad \text{(Violates derivative exponent rule)}$$

1.4 Origins and Engineering Applications

Differential equations naturally model systems undergoing continuous change across science and technology:

- **Population Dynamics:** Malthusian growth models, logistic population growth, and Lotka-Volterra predator-prey systems.
- **Dynamical Systems & Chaos:** Nonlinear pendulum motion, forced oscillators, and atmospheric convection.

- **Fluid Dynamics:** Navier-Stokes equations governing incompressible and compressible flow fields.
- **Heat Transfer:** Heterogeneous conduction, thermal diffusion, and transient cooling.
- **Mathematical Biology & Biophysics:** Enzyme kinetics, nerve impulse propagation (Hodgkin-Huxley model), and epidemic spreads.

1.5 Formation of Differential Equations

A differential equation representing a family of curves is generated by differentiating the general equation containing arbitrary constants and subsequently eliminating those constants.

Worked Example: Elimination of Constants

Form the ordinary differential equation corresponding to the family of curves given by:

$$y = ax + bx^2 \tag{1.9}$$

where a and b are arbitrary constants.
Solution: Since the equation contains two arbitrary constants (a and b), we differentiate Equation (1.9) twice consecutively with respect to x.
Step 1 (First Differentiation):

$$\frac{dy}{dx} = a + 2bx \tag{1.10}$$

Step 2 (Second Differentiation):

$$\frac{d^2y}{dx^2} = 2b \implies b = \frac{1}{2}\frac{d^2y}{dx^2} \tag{1.11}$$

Step 3 (Eliminating Constant a): Substitute the expression for b from Equation (1.11) into Equation (1.10):

$$\frac{dy}{dx} = a + 2\left(\frac{1}{2}\frac{d^2y}{dx^2}\right)x = a + x\frac{d^2y}{dx^2}$$

Solving for a:

$$a = \frac{dy}{dx} - x\frac{d^2y}{dx^2} \tag{1.12}$$

Step 4 (Final Substitution into Original Equation): Substitute a and b back into Equation (1.9):

$$y = \left(\frac{dy}{dx} - x\frac{d^2y}{dx^2}\right)x + \left(\frac{1}{2}\frac{d^2y}{dx^2}\right)x^2$$

Expanding and simplifying:

$$y = x\frac{dy}{dx} - x^2\frac{d^2y}{dx^2} + \frac{1}{2}x^2\frac{d^2y}{dx^2} = x\frac{dy}{dx} - \frac{1}{2}x^2\frac{d^2y}{dx^2}$$

Multiplying through by 2 and rearranging into standard order yields:

$$x^2\frac{d^2y}{dx^2} - 2x\frac{dy}{dx} + 2y = 0 \tag{1.13}$$

This is a 2nd-order, 1st-degree linear ordinary differential equation.

Chapter 2

Solutions and Existence Theory

2.1 General, Particular, and Singular Solutions

Definition: General Solution (Complete Primitive)

The relation involving n independent arbitrary constants that satisfies an n-th order ordinary differential equation is termed its **general solution** (or complete primitive).

Definition: Particular Solution

A **particular solution** is a solution derived from the general solution by assigning specific numerical values to the arbitrary constants, typically dictated by initial or boundary conditions.

Geometric Interpretation

Geometrically, the general solution (primitive) represents an n-parameter **family of curves** satisfying the differential equation, whereas a particular solution represents a **single curve** belonging to this family.

Worked Example: Differential Equation of a Parabolic Family

Form the differential equation of all parabolas whose axes are parallel to the y-axis:

$$(x-h)^2 = 4a(y-k) \tag{2.1}$$

where a, h, k are arbitrary constants.
Solution: Differentiating Equation (2.1) with respect to x:

$$2(x-h) = 4a\frac{dy}{dx} \implies x - h = 2a\frac{dy}{dx}$$

Differentiating a second time with respect to x:

$$1 = 2a\frac{d^2y}{dx^2} \implies \frac{d^2y}{dx^2} = \frac{1}{2a}$$

Differentiating a third time eliminates the constant a:

$$\frac{d^3y}{dx^3} = 0 \tag{2.2}$$

This third-order ordinary differential equation represents the entire family of parabolas.

2.2 Explicit vs. Implicit Solutions

Consider an n-th order real ODE expressed as $F\left(x, y, y', \ldots, y^{(n)}\right) = 0$.

Definition: Explicit Solution

A real-valued function $y = f(x)$ defined on an interval $I \subseteq \mathbb{R}$ is an **explicit solution** if:

1. $f(x)$ possesses continuous derivatives up to order n on I.
2. $F\left(x, f(x), f'(x), \ldots, f^{(n)}(x)\right) = 0$ holds identically for all $x \in I$.

Definition: Implicit Solution

A relation $g(x, y) = 0$ is an **implicit solution** of the ODE on I if the relation implicitly defines at least one function $y = f(x)$ that is an explicit solution on I.

Worked Example: Verification of an Explicit Solution

Show that $f(x) = 2\sin(x) + 3\cos(x)$ is an explicit solution of $y'' + y = 0$ for all $x \in \mathbb{R}$.
Solution: Given $f(x) = 2\sin(x) + 3\cos(x)$, we compute its derivatives:

$$f'(x) = 2\cos(x) - 3\sin(x), \qquad f''(x) = -2\sin(x) - 3\cos(x)$$

Substituting into the differential equation:

$$f''(x) + f(x) = \Big(- 2\sin(x) - 3\cos(x)\Big) + \Big(2\sin(x) + 3\cos(x)\Big) = 0$$

Since this holds identically for all $x \in \mathbb{R}$, $f(x)$ is an explicit solution.

2.3 Initial Value Problems (IVP) and Boundary Value Problems (BVP)

Definition: Initial Value Problem (IVP)

An **Initial Value Problem (IVP)** consists of a differential equation accompanied by supplementary conditions specified at a **single point** x_0:

$$y(x_0) = y_0, \quad y'(x_0) = y_1, \quad \dots, \quad y^{(n-1)}(x_0) = y_{n-1} \tag{2.3}$$

Definition: Boundary Value Problem (BVP)

A **Boundary Value Problem (BVP)** consists of a differential equation accompanied by supplementary conditions specified at **two or more distinct points** x_a, x_b:

$$y(x_a) = y_a, \qquad y(x_b) = y_b \tag{2.4}$$

Worked Example: Contrasting IVP and BVP

Initial Value Problem (IVP):

$$\frac{d^2y}{dx^2} + y = 0, \qquad y(1) = 3, \quad y'(1) = -4 \quad \text{(Both conditions at } x = 1\text{)}$$

Boundary Value Problem (BVP):

$$\frac{d^2y}{dx^2} + y = 0, \qquad y(0) = 1, \quad y\left(\frac{\pi}{2}\right) = 5 \quad \text{(Conditions at } x = 0 \text{ and } x = \pi/2\text{)}$$

2.4 Picard-Lindelöf Existence and Uniqueness Theorem

Theorem: Picard-Lindelöf Theorem

Consider the first-order initial value problem:

$$\frac{dy}{dx} = f(x, y), \qquad y(x_0) = y_0 \tag{2.5}$$

Let $f(x, y)$ and its partial derivative $\frac{\partial f}{\partial y}$ be continuous in a closed rectangular region $D = \{(x, y) \mid |x - x_0| \leq a, |y - y_0| \leq b\}$.
Then, there exists a **unique solution** $y = \phi(x)$ defined on an interval $|x - x_0| \leq h$ (where $h \leq a$), satisfying the initial condition $\phi(x_0) = y_0$.

Chapter 3

First-Order Ordinary Differential Equations

A first-order ODE can be expressed in derivative form $\frac{dy}{dx} = f(x, y)$ or differential form $M(x, y)dx + N(x, y)dy = 0$.

3.1 Separable Equations and Reducible Forms

Definition: Separable Differential Equation

A differential equation is **separable** if it can be written as:

$$F(x)G(y)dx + f(x)g(y)dy = 0 \implies \frac{F(x)}{f(x)}dx + \frac{g(y)}{G(y)}dy = 0 \tag{3.1}$$

Worked Example: Separable Equation Solution

Solve the initial value problem:

$$x \ln(y)dx + (x^2 + 1)\cos(y)dy = 0, \qquad y(1) = \frac{\pi}{2}$$

Solution: Separating variables:

$$\frac{x}{x^2 + 1}dx + \frac{\cos(y)}{\ln(y)}dy = 0$$

For the initial condition $y(1) = \frac{\pi}{2}$, integrating both terms yields the complete explicit curve passing through $(1, \pi/2)$.

Worked Example: Reducible to Separable Form

Solve $\frac{dy}{dx} = (4x + y + 1)^2$.
Solution: Let $u = 4x + y + 1$. Differentiating with respect to x:

$$\frac{du}{dx} = 4 + \frac{dy}{dx} \implies \frac{dy}{dx} = \frac{du}{dx} - 4$$

Substituting into the ODE:

$$\frac{du}{dx} - 4 = u^2 \implies \frac{du}{u^2 + 4} = dx$$

Integrating both sides:

$$\frac{1}{2}\arctan\left(\frac{u}{2}\right) = x + C \implies \arctan\left(\frac{4x + y + 1}{2}\right) = 2x + 2C$$

Taking the tangent of both sides:

$$4x + y + 1 = 2\tan(2x + K) \tag{3.2}$$

3.2 Homogeneous Differential Equations

Definition: Homogeneous Function and DE

A function $f(x, y)$ is homogeneous of degree n if $f(kx, ky) = k^n f(x, y)$. The equation $M(x, y)dx + N(x, y)dy = 0$ is **homogeneous** if M and N are homogeneous functions of the same degree.

Substitution $y = vx$ (hence $\frac{dy}{dx} = v + x\frac{dv}{dx}$) reduces any homogeneous equation into a separable equation.

Worked Example: Homogeneous Differential Equation

Solve $(x^2 + y^2)dx + 2xydy = 0$.
Solution: Expressing in derivative form:

$$\frac{dy}{dx} = -\frac{x^2 + y^2}{2xy}$$

Substituting $y = vx$ and $\frac{dy}{dx} = v + x\frac{dv}{dx}$:

$$v + x\frac{dv}{dx} = -\frac{x^2 + v^2x^2}{2x^2v} = -\frac{1 + v^2}{2v}$$

Rearranging:

$$x\frac{dv}{dx} = -\frac{1 + v^2}{2v} - v = -\frac{1 + 3v^2}{2v} \implies \frac{2v}{1 + 3v^2}dv = -\frac{dx}{x}$$

Integrating both sides:

$$\frac{1}{3}\ln(1 + 3v^2) = -\ln(x) + \ln(C) \implies \ln\left(x^3(1 + 3v^2)\right) = \ln(C')$$

Substituting $v = y/x$:

$$x^3\left(1 + 3\frac{y^2}{x^2}\right) = C \implies x^3 + 3xy^2 = C \tag{3.3}$$

3.3 First-Order Linear Differential Equations

A first-order linear ODE has the canonical form:

$$\frac{dy}{dx} + P(x)y = Q(x) \tag{3.4}$$

Multiplying by the **Integrating Factor (I.F.)** $\mu(x) = e^{\int P(x)dx}$ transforms the left-hand side into an exact product derivative:

$$\frac{d}{dx}\left[ye^{\int P(x)dx}\right] = Q(x)e^{\int P(x)dx} \implies y = e^{-\int P(x)dx}\left[\int Q(x)e^{\int P(x)dx}dx + C\right] \tag{3.5}$$

3.4 Bernoulli Equations

Definition: Bernoulli Equation

An equation of the form:

$$\frac{dy}{dx} + P(x)y = Q(x)y^n \qquad (n \neq 0, 1) \tag{3.6}$$

is a **Bernoulli differential equation**.

Dividing by y^n and substituting $v = y^{1-n}$ transforms it into a first-order linear ODE in v:

$$\frac{dv}{dx} + (1-n)P(x)v = (1-n)Q(x) \tag{3.7}$$

3.5 Exact Differential Equations

Theorem: Condition for Exactness

The differential form $M(x,y)dx + N(x,y)dy = 0$ is **exact** if and only if:

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x} \tag{3.8}$$

If exact, there exists a potential function $F(x,y)$ such that $\frac{\partial F}{\partial x} = M$ and $\frac{\partial F}{\partial y} = N$, giving the general solution $F(x,y) = C$.

3.6 Orthogonal Trajectories

An **orthogonal trajectory** is a curve that intersects every member of a given family of curves at right angles ($90°$).

Rule for Finding Orthogonal Trajectories

1. For Cartesian family $f(x,y,c) = 0$: find ODE $F(x,y,y') = 0$, replace $y' \to -\frac{1}{y'}$, and integrate.
2. For Polar family $f(r,\theta,c) = 0$: find ODE $F(r,\theta,r') = 0$, replace $\frac{dr}{d\theta} \to -r^2\frac{d\theta}{dr}$, and integrate.

Worked Example: Parabolic Orthogonal Trajectories

Find the orthogonal trajectories of the parabolic family $y = cx^2$.
Solution: Differentiating $y = cx^2$ gives $\frac{dy}{dx} = 2cx = 2\left(\frac{y}{x^2}\right)x = \frac{2y}{x}$. Replacing $\frac{dy}{dx}$ with $-\frac{dx}{dy}$ for orthogonality:

$$-\frac{dx}{dy} = \frac{2y}{x} \implies xdx + 2ydy = 0$$

Integrating both sides:

$$\frac{x^2}{2} + y^2 = C' \implies x^2 + 2y^2 = K^2 \tag{3.9}$$

The orthogonal trajectories form a family of **ellipses** centered at the origin.

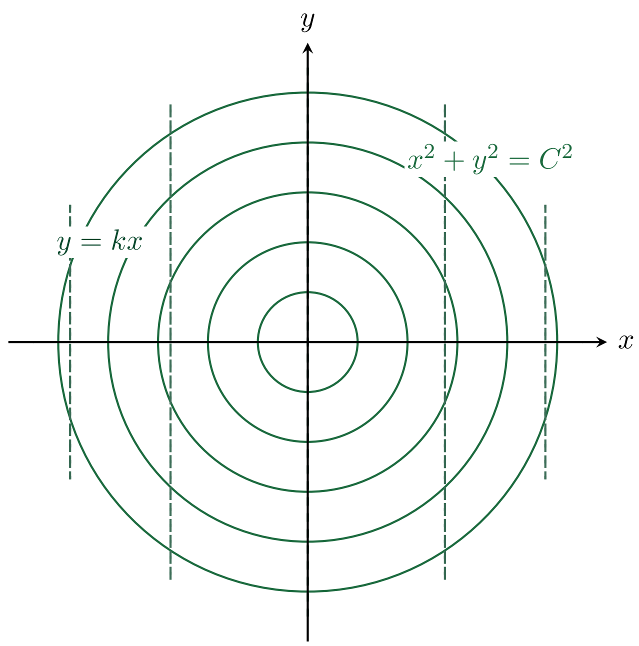

Figure 3.1: Orthogonal trajectories: Concentric circles $x^2+y^2 = C^2$ (solid green) intersected perpendicularly by radial lines $y = kx$ (dashed emerald).

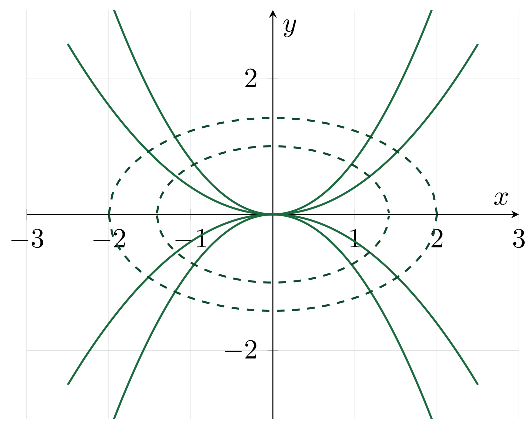

Figure 3.2: Parabolas $y = cx^2$ (solid green) and their orthogonal elliptical trajectories $x^2 + 2y^2 = K^2$ (dashed dark green).

Chapter 4

Higher-Order Linear Differential Equations

4.1 General Theory and Superposition Principle

An n-th order non-homogeneous linear ODE is represented by:

$$a_0(x)\frac{d^n y}{dx^n} + a_1(x)\frac{d^{n-1}y}{dx^{n-1}} + \cdots + a_n(x)y = F(x) \tag{4.1}$$

Theorem: Superposition Principle

If $y_1, y_2, \ldots, y_m$ are solutions of the homogeneous linear ODE ($F(x) = 0$), then any linear combination:

$$y(x) = c_1 y_1(x) + c_2 y_2(x) + \cdots + c_m y_m(x) \tag{4.2}$$

is also a solution.

4.2 Linear Independence and the Wronskian Determinant

Definition: Wronskian Determinant

For n differentiable functions $f_1, f_2, \ldots, f_n$, the **Wronskian** $W(f_1, \ldots, f_n)$ is defined as:

$$W(f_1, f_2, \ldots, f_n)(x) = \begin{vmatrix} f_1(x) & f_2(x) & \ldots & f_n(x) \\ f_1'(x) & f_2'(x) & \ldots & f_n'(x) \\ \vdots & \vdots & \ddots & \vdots \\ f_1^{(n-1)}(x) & f_2^{(n-1)}(x) & \ldots & f_n^{(n-1)}(x) \end{vmatrix} \tag{4.3}$$

Theorem: Wronskian Test for Linear Independence

Solutions $y_1, y_2, \ldots, y_n$ of an n-th order homogeneous linear ODE are **linearly independent** on an interval I if and only if $W(y_1, y_2, \ldots, y_n)(x) \neq 0$ for at least one $x \in I$.

Worked Example: Wronskian Verification

Verify that e^x, e^{-x}, e^{2x} are linearly independent solutions of $y''' - 2y'' - y' + 2y = 0$.
Solution: Evaluating the Wronskian:

$$W(e^x, e^{-x}, e^{2x}) = \begin{vmatrix} e^x & e^{-x} & e^{2x} \\ e^x & -e^{-x} & 2e^{2x} \\ e^x & e^{-x} & 4e^{2x} \end{vmatrix} = e^x \cdot e^{-x} \cdot e^{2x} \begin{vmatrix} 1 & 1 & 1 \\ 1 & -1 & 2 \\ 1 & 1 & 4 \end{vmatrix}$$

Computing the determinant:

$$1(-4-2) - 1(4-2) + 1(1-(-1)) = -6 - 2 + 2 = -6$$

Thus $W = -6e^{2x} \neq 0$ for all $x \in \mathbb{R}$, proving linear independence.

4.3 Reduction of Order

If one non-trivial solution $y_1(x)$ of an n-th order linear homogeneous ODE is known, the substitution $y(x) = v(x)y_1(x)$ reduces the ODE to an $(n-1)$-th order linear ODE in $w = v'$.

4.4 Homogeneous Linear ODEs with Constant Coefficients

For $a_0 y^{(n)} + a_1 y^{(n-1)} + \cdots + a_n y = 0$, substituting $y = e^{mx}$ yields the **Auxiliary (Characteristic) Equation**:

$$a_0 m^n + a_1 m^{n-1} + \cdots + a_n = 0 \tag{4.4}$$

Root Cases for the Auxiliary Equation

1. **Distinct Real Roots** ($m_1 \neq m_2 \ldots \neq m_n$):

$$y_c = c_1 e^{m_1 x} + c_2 e^{m_2 x} + \cdots + c_n e^{m_n x}$$

2. **Repeated Real Roots** ($m_1 = m_2 = \cdots = m_k = m$):

$$y_c = (c_1 + c_2 x + c_3 x^2 + \cdots + c_k x^{k-1}) e^{mx}$$

3. **Complex Conjugate Roots** ($m = \alpha \pm i\beta$):

$$y_c = e^{\alpha x}\Big(c_1 \cos(\beta x) + c_2 \sin(\beta x)\Big)$$

Worked Example: Complex Roots Initial Value Problem

Solve the IVP: $y'' - 6y' + 25y = 0$, with $y(0) = -3, y'(0) = -1$.
Solution: Auxiliary equation: $m^2 - 6m + 25 = 0 \implies m = \frac{6 \pm \sqrt{36-100}}{2} = 3 \pm 4i$. General solution:

$$y(x) = e^{3x}\Big(c_1 \cos(4x) + c_2 \sin(4x)\Big)$$

Applying $y(0) = -3 \implies c_1 = -3$. Differentiating and applying $y'(0) = -1$:

$$y'(x) = 3e^{3x}(c_1 \cos 4x + c_2 \sin 4x) + e^{3x}(-4c_1 \sin 4x + 4c_2 \cos 4x)$$

$$-1 = 3(-3) + 4c_2 \implies 4c_2 = 8 \implies c_2 = 2$$

The unique solution is:

$$y(x) = e^{3x}\Big(2\sin(4x) - 3\cos(4x)\Big) \tag{4.5}$$

4.5 Non-Homogeneous ODEs and Particular Integrals

The general solution of a non-homogeneous ODE is $y = y_c + y_p$ (Complementary Function + Particular Integral).

Using the differential operator $D \equiv \frac{d}{dx}$, $y_p = \frac{1}{f(D)}F(x)$.

Inverse Operator D-Rules

1. **Exponential Source** $F(x) = e^{ax}$:

$$\frac{1}{f(D)}e^{ax} = \frac{1}{f(a)}e^{ax} \quad \text{if } f(a) \neq 0$$

   Failure Case: If $f(a) = 0$, $\frac{1}{f(D)}e^{ax} = x\frac{1}{f'(a)}e^{ax}$.

2. **Sinusoidal Source** $F(x) = \sin(ax)$ or $\cos(ax)$: Replace $D^2 \to -a^2$.

3. **Polynomial Source** $F(x) = x^m$: Expand $[f(D)]^{-1}$ as a binomial series in ascending powers of D.

4. **Exponential Shift**: $\frac{1}{f(D)}\left(e^{ax}V(x)\right) = e^{ax}\frac{1}{f(D+a)}V(x)$.